

Ch9_Data Infrastructure

Data Source

Internal Data Source

企业自身系统产生的数据

1. Master Data

Core business entities such as **customers** (客户), **products** (产品), and **suppliers** (供应商).

- **Master Data (主数据)**
 - **定义:** 企业运营中最核心、最稳定的业务实体信息。
 - **内容:** 客户、产品、供应商、员工等。
 - **作用:** 作为业务交易的“基础字典”，保证数据一致性。
 - **例子:** 客户的姓名、联系方式；产品的型号、规格；供应商的公司信息。

2. Transactional Data

Records of business activities and events, e.g., **orders** (订单), **payments** (付款), **inventory changes** (库存变动).

Useful for operational analysis and financial reporting.

- **Transactional Data (交易数据)**
 - **定义:** 记录企业日常业务活动和事件的数据。
 - **内容:** 订单、发票、付款、库存变动等。
 - **作用:** 反映企业的动态运营情况，通常是大量、持续生成的。
 - **例子:** 某客户下了一笔订单，金额 10,000 元；某产品库存减少 50 件。

3. Metadata (元数据)

Data that describes other data, including **format** (格式), **source** (来源), and **update time** (更新时间).

Critical for **data governance** (数据治理) and tracking.

- **Metadata (元数据)**
 - **定义:** 描述其他数据的属性和结构的信息。
 - **内容:** 数据来源、格式、更新时间、字段定义。
 - **作用:** 帮助理解和管理数据，提升数据可用性。
 - **例子:** 某数据库表的字段说明（如“Customer_ID: 唯一标识客户”）；文件的创建日期和作者。

4. Log Data (日志数据)

Captures user and system actions, such as **clicks** (点击), **system calls** (系统调用), and **error records** (错误记录).

Supports behavior analysis and system monitoring.

- **Log Data (日志数据)**

- **定义:** 记录用户和系统的操作行为。
- **内容:** 登录记录、点击操作、系统错误、访问路径。
- **作用:** 用于安全审计、性能监控、用户行为分析。
- **例子:** 某用户在 2025-11-18 10:00 登录系统; 服务器在 10:05 报错。

External Data Source

来自企业外部的第三方或公共数据

1. Partners

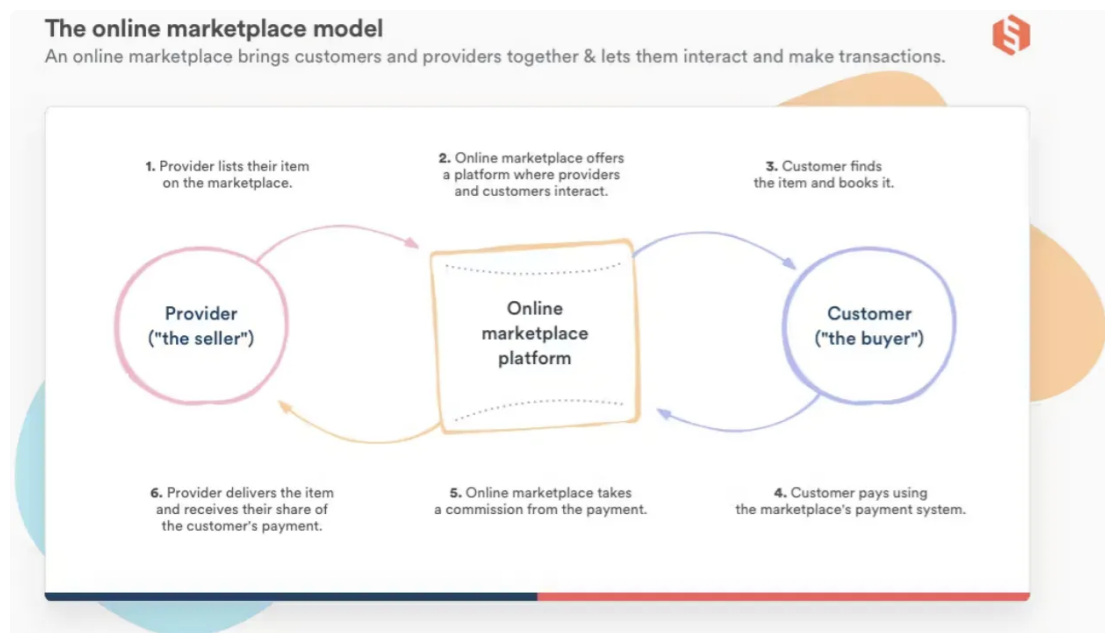
Data from suppliers or customers for collaborative planning.

如, supply chain planning (供应链计划), customer demand forecasting (客户需求预测).

2. Marketplaces (数据交易平台)

An online marketplace is a type of **e-commerce platform** where people exchange goods, services, or information.

- Marketplaces have a multi-sided model with **at least two distinct user groups** (e.g., buyers and sellers, renters and providers).
- They **facilitate transactions** between the user groups.
- Customers can take actions like listing, browsing, booking, and purchasing directly on the platform.
- Marketplaces **do not usually own the inventory** themselves.



Based on participant involvement, we have:

- C2C (Consumer-to-Consumer) marketplaces

- B2C (Business-to-Consumer) marketplaces
- B2B (Business-to-Business) marketplaces

Platforms to buy or sell data, providing insights into **market trends (市场趋势)** and **customer behavior (消费者行为)**.

Often used for competitive intelligence and strategic planning.

- **Competitive Intelligence:** The process of collecting and analyzing information about competitors, markets, and industry trends to support business decisions.
- **Strategic Planning:** The long-term process of defining goals, setting priorities, and allocating resources to achieve sustainable growth.

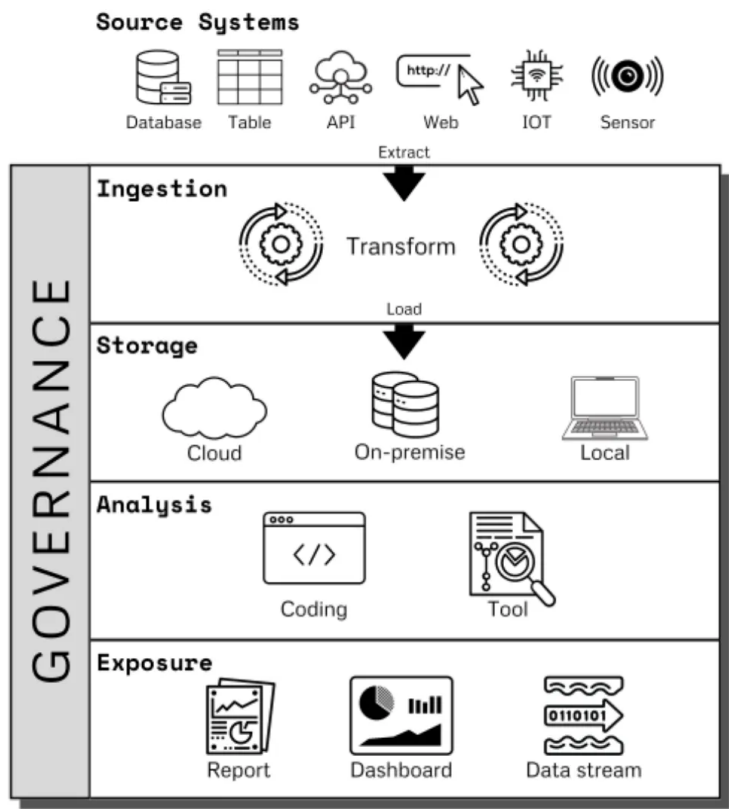
Category	Advantage	Disadvantage
Access to diverse data	Enables access to a wide range of datasets across industries and domains 可获取跨行业、多领域的数据资源	May include inconsistent or poorly documented data 数据可能缺乏一致性或文档不完整
Cost efficiency	Reduces the need to build proprietary datasets from scratch 降低自建数据成本	High-quality data may be expensive or locked behind licensing 高质量数据可能价格昂贵或受限于许可
Speed and agility	Accelerates time-to-insight for analytics and decision-making 快速获取分析洞察，加快决策	Risk of outdated or irrelevant data 存在数据过时或不相关的风险
Scalability	Supports large-scale data acquisition for AI and analytics 支持大规模数据获取，适用于 AI 与分析	Integration with internal systems can be complex 与内部系统集成可能复杂
Third-party enrichment	Enhances internal data with external context (e.g., market trends, demographics) 可用外部数据丰富内部分析 (如市场趋势、人口统计)	Potential privacy and compliance risks 存在隐私与合规风险

3. Public Sources (公共数据来源)

Data provided by governments or NGOs, such as **census (人口普查)** and **economic indicators (经济指标)**.

Useful for macro-level analysis and policy research.

Data System Architecture



1. Source Systems
2. Ingestion
 - 提取（Extract）：从数据源抓取原始数据
 - 转换（Transform）：对数据进行清洗、格式转换、标准化等处理（如字段重命名、单位统一）
3. Storage

Data Warehouse	数据仓库	面向结构化分析的集中式数据平台
Data Lake	数据湖	存储原始数据的灵活平台，支持多种格式
Data Mesh	数据网格	分布式、以业务域为中心的数据架构，强调团队自治与协作

Data Warehouse	High performance and scalability 性能强，支持扩展 Supports concurrent users 可支持多用户并发访问	Expensive to maintain and update 维护成本高 Limited flexibility for new or changing data formats 对新数据源或格式适应性差
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Data Lake	Flexible for any type of data 支持任意数据类型 Low-cost, scalable storage 存储成本低, 易扩展 Compatible with multiple tools 可与多种分析工具兼容	Low data quality and reliability 数据质量与可靠性低 Poor performance 性能较差 Difficult to locate relevant data 数据查找困难
Data Mesh	Agile, decentralized data sharing 敏捷的分布式数据共享 High data quality and interoperability 数据质量高, 跨域兼容性强	Complex to implement and manage 实施与管理复杂 High overhead and duplication 资源消耗大, 数据重复 Requires strong cross-team coordination 需跨团队高度协作

4. Analysis

分析阶段是对存储的数据进行处理与洞察：

- **编程分析 (Coding)**：使用 Python、SQL 等语言进行数据建模、统计分析
- **工具分析 (Tool)**：使用 BI 工具（如 Power BI、Tableau）进行可视化与业务分析

5. Exposure 数据呈现

- **报告 (Report)**：生成 PDF、Word 等格式的分析文档
- **仪表板 (Dashboard)**：实时可视化界面，展示关键指标
- **数据流 (Data Stream)**：将数据实时推送给其他系统或服务，如 API 输出或消息队列

【Governance 治理层贯穿全流程】

治理是整个数据管道的核心保障，涵盖：

- **数据质量管理**：确保数据准确、完整、一致 accuracy, completeness, and consistency
- **安全与隐私保护**：控制访问权限，防止泄露 Control access rights and prevent unauthorized use or leakage
- **合规性 (Compliance)**：遵守 GDPR、PIPL 等法规
- **元数据管理**：记录数据来源、变更历史、使用情况 Track data origin, transformation history, and usage context
- **责任归属 (Accountability)**：明确谁负责数据的创建、维护与使用

